Advanced Statistical Methods II

Tutorial II: Imputation and EM Computations in R

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Outline of Today's Tutorial

Load the reproducible simulated data

The data file used in these slides was generated from the simple regression-like population

$$Y = 2 + 1.2X + \varepsilon, \qquad \varepsilon \sim N(0, 0.8^2).$$

```
imp_data <- read.csv("tutorial23_imputation_data.csv")

x      <- imp_data$x
y      <- imp_data$y
p_miss <- imp_data$p_miss
R      <- imp_data$R
y_obs  <- imp_data$y_obs

dat <- data.frame(x = x, y = y_obs, R = R)
complete_data <- data.frame(x = x, y = y)
summary(complete_data)
```

Create a MAR-style missingness pattern

Let the probability that Y is missing depend on observed X:

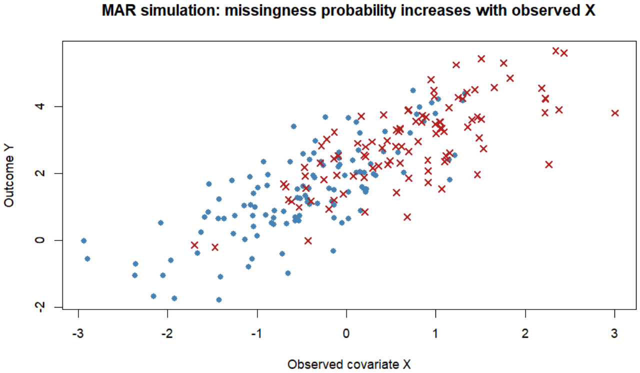
$$\mathbb{P}(R_i = 1 \mid X_i, Y_i) = \mathbb{P}(R_i = 1 \mid X_i) = \frac{\exp(-0.3 + 1.4X_i)}{1 + \exp(-0.3 + 1.4X_i)}.$$

Here $R_i = 1$ means Y_i is missing.

```
# The data file stores one reproducible realization from this MAR rule.

# y_obs is the observed version of y.
y_obs[R == 1][1:5]
mean(is.na(dat$y)) # .459
```

Visualizing the missingness pattern



Three quick imputations

We compare three single-imputation rules.

```
imp_out <- read.csv("tutorial23_imputation_outputs.csv")
mean_imp  <- imp_out$mean_imp
reg_imp   <- imp_out$reg_imp
stoch_imp <- imp_out$stoch_imp
mu_hat    <- imp_out$mu_hat

fit_obs <- lm(y_obs ~ x, na.action = na.exclude)
sigma_hat <- summary(fit_obs)$sigma
coef(fit_obs)

# (Intercept)           x
#   2.005699    1.198375

sigma_hat # 0.9006541
```

What changes after single imputation?

Single imputation treats guessed values as if they were observed. This usually makes downstream standard errors too small; see van Buuren (2018, Ch. 4) and Little and Rubin (2019, Ch. 4).

What does “downstream standard errors too small” mean?

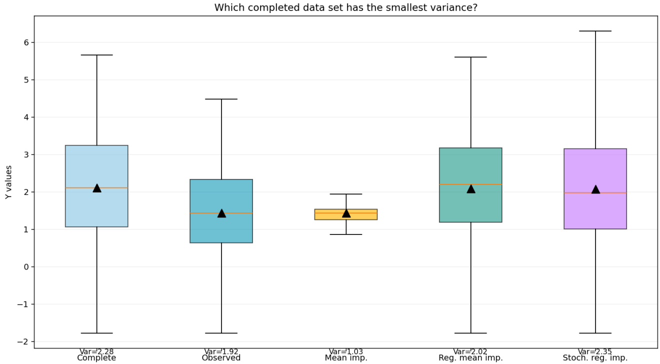
If we later run `lm()` or a test on the filled data, the software treats the imputed values as real observations. It then ignores two extra sources of uncertainty: the uncertainty in estimating the imputation model and the residual uncertainty in the missing Y's. The resulting standard errors can therefore be too optimistic.

A compact comparison

```r
compare <- rbind(
  complete    = c(mean(y), var(y), coef(lm(y ~ x))[2]),
  observed    = c(mean(y_obs, na.rm = TRUE), var(y_obs, na.rm = TRUE),
                  coef(lm(y_obs ~ x))[2]),
  mean_imp    = c(mean(mean_imp), var(mean_imp), coef(lm(mean_imp ~ x))[2]),
  reg_imp     = c(mean(reg_imp), var(reg_imp), coef(lm(reg_imp ~ x))[2]),
  stoch_imp   = c(mean(stoch_imp), var(stoch_imp), coef(lm(stoch_imp ~ x))[2])
)
colnames(compare) <- c("mean", "variance", "slope")
round(compare, 3)
```

Which method will give the smallest variance of imputed Y. Why?

Which imputation has the smallest variance?



Mean imputation usually gives the smallest variance because every missing value is replaced by the same number. Regression mean imputation also smooths the missing values onto a fitted line. Stochastic regression imputation restores residual variability by adding random draws around the fitted conditional mean.

Bivariate normal conditional imputation

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathrm{N}_2 \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_{XX} & \sigma_{XY} \\ \sigma_{XY} & \sigma_{YY} \end{pmatrix} \right).$$

If Y is missing but $X = x$ is observed, then

$$Y \mid X = x \sim \mathrm{N} \left(\mu_Y + \frac{\sigma_{XY}}{\sigma_{XX}}(x - \mu_X), \sigma_{YY} - \frac{\sigma_{XY}^2}{\sigma_{XX}} \right).$$

Conditional mean imputation uses the first quantity. Stochastic imputation also uses the conditional variance.

Estimate the conditional distribution in R

For a complete or partially observed bivariate sample, estimate the regression of Y on X using the complete pairs.

```
obs <- !is.na(dat$y)
fit_y_on_x <- lm(y ~ x, data = dat, subset = obs)

alpha_hat <- coef(fit_y_on_x)[1]
beta_hat  <- coef(fit_y_on_x)[2]
sigma_hat <- summary(fit_y_on_x)$sigma

c(alpha = alpha_hat, beta = beta_hat, sigma = sigma_hat)
```

For bivariate normal data, the regression line equals the conditional mean:

$$\mathbb{E}(Y \mid X = x) = \alpha + \beta x.$$

Conditional mean vs conditional draw

```
mu_cond <- predict(fit_y_on_x, newdata = dat)

# Conditional mean imputation
imp_cond_mean <- y_obs
imp_cond_mean[is.na(y_obs)] <- mu_cond[is.na(y_obs)]

# Conditional stochastic imputation
imp_cond_draw <- y_obs
imp_cond_draw[is.na(y_obs)] <- mu_cond[is.na(y_obs)] +
  rnorm(sum(is.na(y_obs)), mean = 0, sd = sigma_hat)
```

Interpretation

For a missing observation with large X_i, conditional imputation does not borrow the overall mean. It borrows from similar values of X.

Do the two imputations target the same object?

For a missing Y_i, compare

$$\widehat{y}_i^{\text{mean}} = \widehat{\mathbb{E}}(Y_i \mid X_i)$$

and

$$\widehat{y}_i^{\text{draw}} = \widehat{\mathbb{E}}(Y_i \mid X_i) + e_i, \qquad e_i \sim N(0, \widehat{\sigma}^2).$$

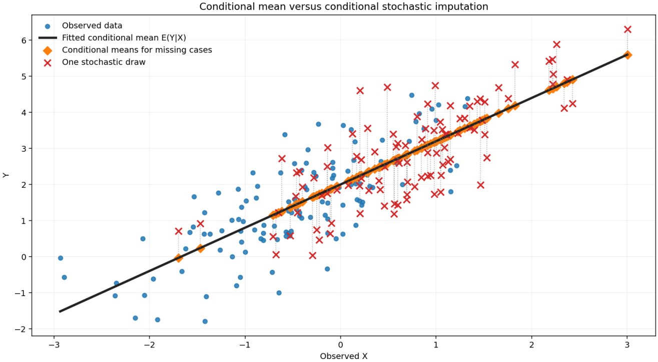
Which one is better if your goal is prediction? Which one is better if your goal is to represent population variability?

Solution

- For predicting the conditional mean, $\widehat{\mathbb{E}}(Y_i \mid X_i)$ is usually more stable.
- For representing the distribution of plausible missing values, a conditional draw is more realistic.
- The conditional draw adds residual uncertainty; the conditional mean does not.

Conditional mean imputation gives a smooth fitted value. Conditional stochastic imputation gives a plausible unseen value. These are different inferential targets.

Solution pictorially



The diamond markers lie on the fitted conditional mean curve. The crosses are one possible stochastic completion. The vertical gaps represent residual uncertainty that the conditional mean alone suppresses.

Now let (X_i, Y_i) be bivariate normal, and suppose some Y_i's are missing while all X_i's are observed.

- Complete-data sufficient statistics involve

$$\sum X_i, \quad \sum Y_i, \quad \sum X_i^2, \quad \sum X_i Y_i, \quad \sum Y_i^2.$$

- When Y_i is missing, we replace missing sufficient statistics by their conditional expectations.

E-step quantities

At iteration t, write

$$\theta^{(t)} = (\mu_X^{(t)}, \mu_Y^{(t)}, \sigma_{XX}^{(t)}, \sigma_{XY}^{(t)}, \sigma_{YY}^{(t)}).$$

For each missing Y_i, compute

$$m_i^{(t)} = \mathbb{E}_{\theta^{(t)}}(Y_i \mid X_i = x_i) = \mu_Y^{(t)} + \frac{\sigma_{XY}^{(t)}}{\sigma_{XX}^{(t)}}(x_i - \mu_X^{(t)}),$$

$$v^{(t)} = \mathbb{V}\mathrm{ar}_{\theta^{(t)}}(Y_i \mid X_i = x_i) = \sigma_{YY}^{(t)} - \frac{(\sigma_{XY}^{(t)})^2}{\sigma_{XX}^{(t)}}.$$

Then

$$\mathbb{E}(Y_i^2 \mid X_i = x_i) = v^{(t)} + \{m_i^{(t)}\}^2.$$

M-step quantities

Construct expected complete-data sums:

$$S_Y^{(t)} = \sum_{i:Y_i\ obs} y_i + \sum_{i:Y_i\ mis} m_i^{(t)},$$

$$S_{XY}^{(t)} = \sum_{i:Y_i\ obs} x_i y_i + \sum_{i:Y_i\ mis} x_i m_i^{(t)},$$

$$S_{YY}^{(t)} = \sum_{i:Y_i\ obs} y_i^2 + \sum_{i:Y_i\ mis} \{v^{(t)} + (m_i^{(t)})^2\}.$$

Update μ and Σ as the usual complete-data MLEs, but using these expected sums.

Do not update σ_{YY} using only $(m_i^{(t)})^2$. The conditional variance term $v^{(t)}$ is essential.

Find the hidden bug

A student proposes this line in the M-step:

$$S_{YY}^{(t)} = \sum_{obs} y_i^2 + \sum_{mis} \{m_i^{(t)}\}^2.$$

What is wrong with this update? What qualitative bias might it create in $\widehat{\sigma}_{YY}$?

Solution

The update ignores conditional variance:

$$\mathbb{E}(Y_i^2 \mid X_i = x_i) \neq \{\mathbb{E}(Y_i \mid X_i = x_i)\}^2.$$

The correct identity is

$$\mathbb{E}(Y_i^2 \mid X_i = x_i) = \mathbb{V}\mathrm{ar}(Y_i \mid X_i = x_i) + \{\mathbb{E}(Y_i \mid X_i = x_i)\}^2.$$

Leaving out $v^{(t)}$ treats every imputed value as perfectly known. It generally underestimates variance.

Mixture model as a missing-label problem

For observations $X_1, \ldots, X_n$, suppose

$$f(x_i \mid \theta) = \pi\phi(x_i; \mu_1, \sigma_1^2) + (1-\pi)\phi(x_i; \mu_2, \sigma_2^2).$$

Introduce latent labels

$$Z_i = \begin{cases} 1, & \text{component } 1, \\ 0, & \text{component } 2. \end{cases}$$

Then the complete data would be (X_i, Z_i). This is the standard missing-label view of finite mixtures; see McLachlan and Peel (2000) and McLachlan and Krishnan (2008, Ch. 3).

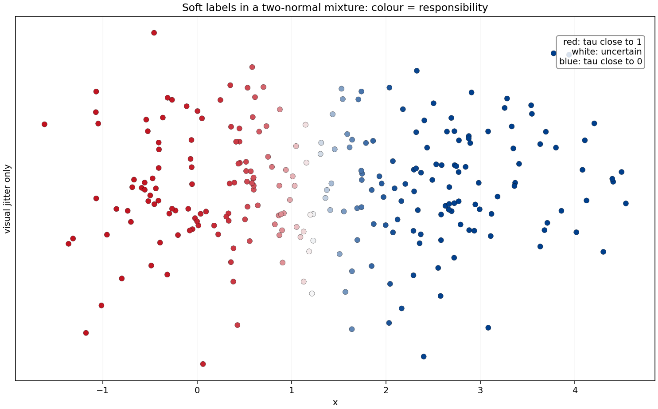
E-step: responsibilities

At iteration t, compute the posterior probability that observation i belongs to component 1:

$$\tau_i^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = 1 \mid X_i = x_i)$$

$$= \frac{\pi^{(t)}\phi(x_i; \mu_1^{(t)}, (\sigma_1^2)^{(t)})}{\pi^{(t)}\phi(x_i; \mu_1^{(t)}, (\sigma_1^2)^{(t)}) + (1 - \pi^{(t)})\phi(x_i; \mu_2^{(t)}, (\sigma_2^2)^{(t)})}.$$

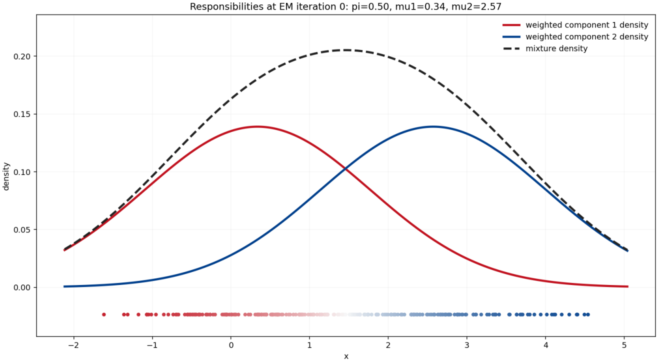
$\tau_i^{(t)}$ is a soft label. It is the EM replacement for the unobserved hard label Z_i.



Interpretation

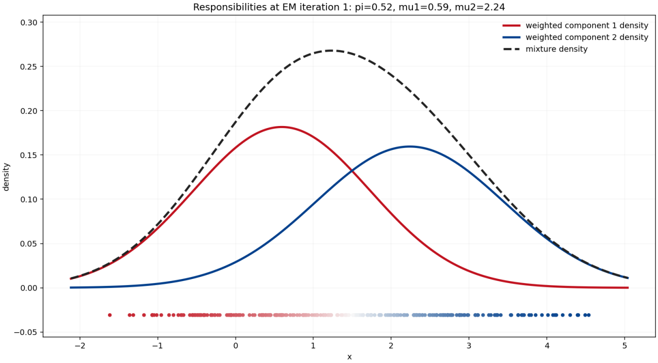
Points in the overlap region are not forced into one cluster. EM assigns fractional membership through responsibilities.

Responsibilities after EM iteration 0

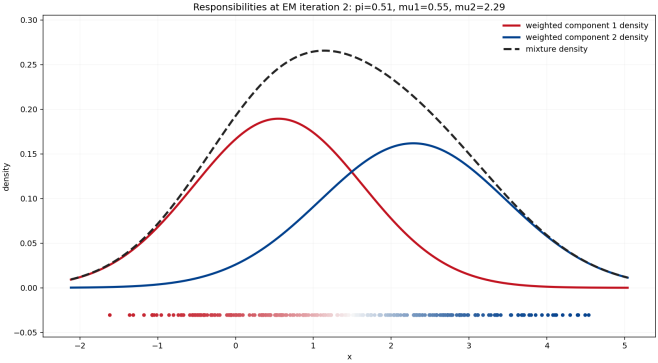


Each point is colored by $\tau_i^{(0)}$, the current posterior membership probability for component 1. Watch how the uncertain overlap region changes more slowly than the clear left and right regions.

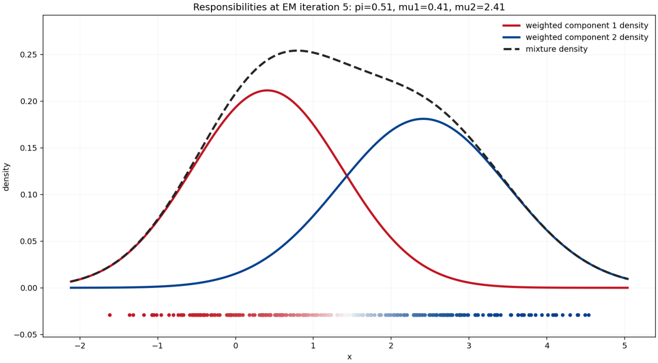
Responsibilities after EM iteration 1



Responsibilities after EM iteration 2



Responsibilities after EM iteration 5



M-step: weighted complete-data MLEs

With responsibilities $\tau_i^{(t)}$, the updates are

$$\pi^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n} \tau_i^{(t)},$$

$$\mu_1^{(t+1)} = \frac{\sum_i \tau_i^{(t)} x_i}{\sum_i \tau_i^{(t)}}, \qquad \mu_2^{(t+1)} = \frac{\sum_i (1-\tau_i^{(t)}) x_i}{\sum_i (1-\tau_i^{(t)})},$$

$$(\sigma_1^2)^{(t+1)} = \frac{\sum_i \tau_i^{(t)} (x_i - \mu_1^{(t+1)})^2}{\sum_i \tau_i^{(t)}},$$

$$(\sigma_2^2)^{(t+1)} = \frac{\sum_i (1-\tau_i^{(t)}) (x_i - \mu_2^{(t+1)})^2}{\sum_i (1-\tau_i^{(t)})}.$$

Load a reproducible two-normal mixture

```r
mix_data <- read.csv("tutorial23_mixture_data.csv")
xmix <- mix_data$xmix
z_true <- mix_data$z_true   # available only because simulated

hist(xmix, breaks = 25, col = "lightgray", border = "white",
     main = "Simulated two-normal mixture", xlab = "x")
```

In a real dataset, z would not be observed. EM treats z as the missing data.

Implementation issues

Useful prompts:

- What happens if one component gets almost no points?
- What happens if σ_k becomes extremely small?
- What happens if the initial values put both means close together?

Issues

1. **Component collapse:** if $\sum_i \tau_i$ is almost zero, the updates for one component become unstable.
2. **Variance collapse:** a component may concentrate around one point, driving $\widehat{\sigma}_k$ toward zero and producing a degenerate likelihood behavior.
3. **Poor starts:** if the starting means are too close, EM may converge slowly or to an unhelpful local maximum.

Three mixture-EM traps: a diagnostic table

Issue	What you may see	Practical response
Component collapse	$\sum_i \tau_i$ nearly zero for a component	restart, constrain, or reduce number of components
Variance collapse	$\widehat{\sigma}_k$ becomes extremely small	use lower bounds/regularization; inspect single-point clusters
Poor starts	slow convergence or inferior local maximum	try multiple starts and compare log-likelihoods

The ascent property protects the likelihood sequence for a given run. It does not protect against bad modelling choices, degeneracy, or a poor local solution.

Label switching

Suppose one EM run returns

$$(\widehat{\pi}, \widehat{\mu}_1, \widehat{\sigma}_1, \widehat{\mu}_2, \widehat{\sigma}_2) = (0.42, 0.05, 0.62, 2.35, 0.81),$$

and another returns

$$(0.58, 2.35, 0.81, 0.05, 0.62).$$

Are these two fitted mixture distributions different?

They are not different as mixture distributions. They are the same model with component labels swapped.

$$\pi\phi(x;\mu_1,\sigma_1^2) + (1-\pi)\phi(x;\mu_2,\sigma_2^2)$$

can be rewritten as

$$(1-\pi)\phi(x;\mu_2,\sigma_2^2) + \pi\phi(x;\mu_1,\sigma_1^2).$$

The labels “component 1” and “component 2” are artificial unless the scientific problem gives them meaning.

Checklist for an EM implementation

1. Write the complete-data log-likelihood.
2. Identify missing sufficient statistics.
3. Compute conditional expectations in the E-step.
4. Use weighted or expected complete-data MLEs in the M-step.
5. Track the observed log-likelihood.
6. Try multiple starting values.
7. Check whether parameter estimates are scientifically interpretable.

Do not judge an EM algorithm only by convergence of parameters. Also examine the log-likelihood and fitted model.

Suggested practice after class

1. Modify the MAR mechanism so that missingness depends less strongly on X. What changes?
2. Repeat the imputation comparison under MCAR and MNAR-style simulations.
3. For the mixture model, try poor starting values and inspect the likelihood trace.
4. Extend the mixture EM code to a three-component normal mixture.

References

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